Whether Small Transformers Copy a Feature from Context Depends on the Noun's Training History

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Abstract

A language model can predict a word's properties by copying them from the preceding text or by recalling what it has stored about the word. We ask what in a single word's training history decides which. We trained 150 small transformers on an artificial language describing objects by shape, colour and texture, and controlled how often 40 new nouns occurred, with which shapes, and how their sentences were arranged. Never-seen nouns copied a shape from a preceding sentence, but mainly because their input embeddings were untrained. Nouns met one sentence at a time stopped copying within about 100 occurrences, whatever they had been seen with. Nouns met in short bursts that shared a shape, with the shape changing between bursts, kept copying: 3.4 nats more than the same sentences spaced out (pre-registered) and 2.9 nats more after 1,000 occurrences than bursts with a fixed shape (exploratory). A stored shape did not itself reduce copying, and the copying was not tied to the noun. In these models, copying persisted where a word's own earlier mentions had been the only source of its current shape. In pretrained Pythia models, replacing nouns' embeddings had no consistent effect.

1 Introduction

When a language model reads *The car was red. The car was*, it can predict *red* because the context says so, or predict a typical car colour from what it has stored. Which source it relies on matters for how models handle documents that contradict their knowledge (Longpre et al., 2021; Xie et al., 2024), for how they learn new words and tokens, and for how we interpret probes that use invented words. Work on in-context and in-weights learning shows that the balance depends on the training data. Classes that occur in bursts, or whose meaning changes, favour learning from context; common, stable classes favour learning stored in the weights (Chan et al., 2022, 2025). Inside pretrained models, mechanisms that copy from the context compete with mechanisms that recall stored facts (Yu et al., 2023; Jin et al., 2024; Ortu et al., 2024). These results describe populations of items. We ask the same question about a single word: *what in a word's own training history decides whether a model copies its features from the context?*

Three accounts are plausible, and they make different predictions.

- **Memory competition.** A model copies until it has stored something about the word; stored knowledge then competes with the context. Words with a stored feature should stop copying sooner than words without one.
- **Representation.** Copying for new words reflects their unusual, untrained embeddings. Once a word's embedding looks like those of familiar words, copying should stop, whatever the word was seen with.
- **Useful context.** A model learns, word by word, whether copying has paid off. A word should keep being copied if, in training, its own earlier mentions predicted its features and memory could not.

Natural data cannot separate these, because frequent words are both familiar and well known. Invented (nonce) words, the usual probe (Thrush et al., 2020; Misra et al., 2023), are new in both respects at once, and conclusions drawn with them can depend on how the new word is represented (Kim et al., 2022; Cacioli, 2026).

We therefore built the separation into training. Small transformers learned an artificial language describing objects by shape, colour and texture, into which we inserted 40 new nouns. For each noun we set how often it occurred (from never to 10,000 sentences), which shapes it occurred with, and whether its sentences came one at a time or in short bursts (Figure 1). We measured how strongly each model copied a shape named in one sentence

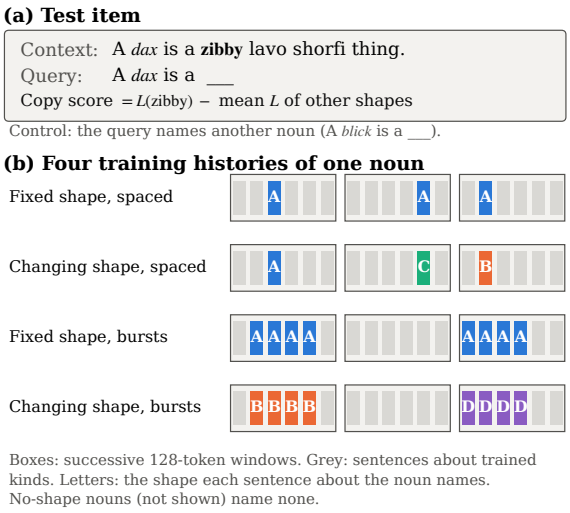


Figure 1: (a) A test item and the copy score. (b) Four of the training histories: each box is a training window of 128 tokens; letters are the shape named by a sentence about the noun.

into the next mention of the noun. A first pre-registered experiment tested the memory and representation accounts; its results pointed to the useful-context account, which a second pre-registered experiment tested. We then asked whether the effect of a word's representation carries over to pretrained models.

The results favour the useful-context account. New nouns copied mainly because of their untrained embeddings. Familiar nouns met one sentence at a time stopped copying quickly, whether or not they had a stored shape. Only nouns whose earlier mentions had been the sole source of their current shape kept copying, and what they kept was a noun-specific tendency to copy, not a link between the noun and its shape. Our claims are about transformers, although we note parallels with children's word learning in Section 5.

2 Related work

In-context and in-weights learning. In-context learning emerges when items occur in bursts, more when their meanings change, and trades off against learning in the weights (Chan et al., 2022); it depends on task diversity (Raventós et al., 2023), favours rare classes and can fade as stored solutions take over (Chan et al., 2025; Singh et al., 2023). Its onset has been traced to induction heads (Olsson et al., 2022; Reddy, 2024), and in small transformers stored bigram statistics are learned quickly and in-context ones later, through an induction head (Bietti et al., 2023). Resetting embeddings during training keeps in-context use alive for rare tokens (Anand et al., 2025), and in synthetic biographies frequent entities shift from context to stored knowledge (Kim et al., 2026). We manipulate frequency, stored knowledge and arrangement separately for single words within one model.

Copying and recall. Language models have long gained from copying recent words, through caches and pointers (Kuhn and De Mori, 1990; Grave et al., 2017; Merity et al., 2017), because words recur in bursts: a word that has occurred once is far more likely to occur again (Church and Gale, 1995; Church, 2000), a regularity that human memory also tracks (Anderson and Schooler, 1991). Pretrained models answer from memory more for frequent entities (Mallen et al., 2023; Kandpal et al., 2023; Razeghi et al., 2022) and resist contexts that contradict what they know (Longpre et al., 2021; Xie et al., 2024; Du et al., 2024); copying and recall compete inside the network (Yu et al., 2023; Jin et al., 2024; Ortu et al., 2024).

New words and new tokens. Rare and new words have poor representations (Schick and Schütze, 2020), under-trained tokens behave abnormally (Land and Bartolo, 2024), and new embeddings are often initialised at the mean of existing ones (Hewitt, 2021). Lexical exposure can inflate apparent generalisation (Kim et al., 2022), and when language models acquire a word depends mainly on its frequency (Chang and Bergen, 2022). Children's word learning offers related evidence on shape (Landau et al., 1988; Smith et al., 2002), on expectations about how features vary (Kemp et al., 2007; Dewar and Xu, 2010) and on the timing of exposures (Childers and Tomasello, 2002; Schwab and Lew-Williams, 2016), to which we return in Section 5.

3 Setup

Language. Each corpus describes 32 kinds of object, each named by an invented noun (*wug*, *dax*, . . .), with 14 exemplars per kind. An exemplar has a shape, a colour and a texture, each one of ten invented words, and is described in sentences such as *A wug is a zibby lavo shorfi thing* (shape, colour, texture) or *Look at the wug. It is zibby and lavo*. In **stable-shape** corpora each kind keeps one shape, so a kind's shape can be stored. In **shifting-shape** corpora a kind gets a new shape in every pass through the training text and keeps it within

History	Sentences about the noun	What it isolates
Unseen	none	an untrained embedding
Fixed shape	always its own shape	a stored shape
Changing shape	a new shape each sentence (spaced) or each burst	a shape slot, nothing to store
No shape	never a shape	familiarity alone
Spaced vs bursts	one sentence per window vs four	recurrence in context

Table 1: Noun histories. In changing-shape bursts the shape is shared within a burst and changes between bursts, so the noun's own earlier sentence is the only source of its current shape.

the pass, so its current shape can come only from the context. In both, sentences about a kind come in runs of four; a control corpus uses random order. Every word is one token.

Models. Decoder-only transformers with pre-layer normalisation and learned position embeddings, with 3.4M, 11.0M and 25.6M parameters (4 to 8 layers), read the text once in non-overlapping windows of 128 tokens. Input and output embeddings are untied, so a noun's input embedding (its *row*) carries everything the model has learned about the noun as an input. We trained ten seeds of each size (Appendix C).

Controlling a noun's history. We added 40 new nouns to each corpus. Four were never used in training (*unseen*). Each of the others had one history (Table 1) at a set number of exposures, counted in sentences, with three nouns per history and exposure level, rotated over nouns across seeds. A noun's sentences were spread evenly from 5% to 95% of training, and each lay wholly inside one window. Presented *spaced*, a noun has one sentence per window; presented in *bursts*, four consecutive sentences in one window. Every exposure was verified from the tokens, and the corpora matched checksums recorded before training.

Two experiments. Experiment 1 (90 models) crossed fixed-, changing- and no-shape nouns, all spaced, with 10, 100, 1,000 and 10,000 exposures, in stable-shape, shifting-shape and random-order corpora. Fixed- and changing-shape nouns appeared in five sentence frames, and no-shape nouns in a frame that names no shape (*The noun is colour and texture*). Experiment 2 (60 models) crossed the same three histories with spaced and burst presentation at 100 and 1,000 exposures, in stable-shape and shifting-shape corpora. To match frames, fixed- and changing-shape nouns appeared only in the test frame, and no-shape nouns in that frame without the shape (*A noun is a colour texture thing*). The experiments also differ in how much of the text the new nouns fill (62% and 8% of windows), so we compare conditions only within an experiment.

Measures. A test item gives one sentence about a noun naming a shape s, then the start of the next mention, *A noun is a*, and reads the model's log probability L of each of the ten shape words at the next position (Figure 1a). The **copy score** is $L(s)$ minus the mean of L over the other shapes, leaving out the noun's own shape if it has one; s is never that shape. It is in nats: a score of 0 means the context's shape is no more likely than the others, and 3.4 means it is about 30 times as likely as the (geometric) average other shape. Each noun gets 36 items. In a **control**, the query names a different noun with the same history; if copying is tied to the noun, the control score should be lower. **Knowledge** is the preference for a noun's own shape after an unrelated sentence. We also score unseen nouns with a **typical embedding**: the mean of the trained kinds' rows plus a random offset of typical length.

Analysis and registration. The seed is the unit of replication: we average each condition over the three model sizes within a seed and test across ten seeds, with Holm correction within each experiment, and count a prediction as supported only if its direction also holds in two of the three sizes. Both experiments were registered before any model was trained (Experiment 1 before its corpora were built). Below, ^{P} marks pre-registered tests and ^{E} exploratory analyses. Appendix G maps every claim to its registered hypothesis and lists all registered tests, including those not discussed here.

4 Results

4.1 New nouns copy because their embeddings are untrained

In stable-shape models of Experiment 1, unseen nouns had a copy score of 4.53 nats [95% CI 3.54, 5.52], but −0.33 when given a typical embedding (Figure 2, left). Earlier experiments in the same language had found the same: arranging text in runs raised an index of copying for unseen nouns by 11.08 nats [9.57, 12.59] over random order, and by only 3.00 [1.79, 4.21] with typical embeddings (Appendix F). To see what about an unseen embed-

ding matters, we changed only the rows of unseen nouns in the Experiment 1 models. Scaling a row to the trained kinds' size, keeping its direction, lowered the score from 4.53 to 4.00 (difference 0.52 [0.20, 0.85]^{P}), almost entirely in the smallest models. A random direction of the same size gave the same score (4.09^{E}), but the trained kinds' mean row removed copying (-0.44^{E}). Copying for unseen nouns thus depends on the row lying away from where trained rows lie. Probes with new words therefore measure the new word's representation as much as anything learned about context.

4.2 Nouns met one at a time stop copying, whatever they store

Experiment 1 tested the memory account. Fixed-shape nouns stored their shape: their preference for it after an unrelated sentence reached 8.3 to 9.5 nats at 1,000 exposures in the three corpora^{P}. If stored knowledge drives copying out, their copy score should have fallen faster than that of no-shape nouns. It did not^{P} (Figure 2, left). In stable-shape models every familiar noun lost copying at the same pace: 3.6 to 4.5 nats at 10 exposures, 0.2 to 0.9 at 100, and within 0.4 nats of zero from 1,000 on. Fixed-shape nouns stayed within 0.4 nats of changing-shape nouns at every exposure^{E}, and no-shape nouns at 1,000 and 10,000 exposures ended close to unseen nouns with a typical embedding (0.33 nats [0.13, 0.52] above them, a reliable difference that is still within the registered 1-nat equivalence bound^{P}). In random-order corpora no noun copied^{E}, so copying is learned from the runs in the text.

This fits the representation account: once a noun was familiar, it stopped copying whether or not it had a stored shape. But in Experiment 1 a familiar noun never recurred within a window, whereas the trained kinds did, and no-shape nouns never appeared in the test frame. Either could explain the loss of copying. Experiment 2 removed both differences.

4.3 Bursts with changing shapes keep copying alive

In Experiment 2 the same histories came either spaced or in bursts. The useful-context account predicts that only changing-shape bursts, where a noun's previous sentence gives its current shape and memory cannot, keep copying. That is what we found (Figure 3, left; Table 2). Changing-shape bursts scored 3.39 nats [2.69, 4.09] more than changing-shape spaced nouns^{P}, as much at 1,000 exposures (3.35) as at 100 (3.43). After 1,000 exposures they still scored 3.85 nats, against 0.5 to 1.0 for every other familiar history.

Comparison (stable shapes)	Pooled	1,000 exp.
Changing: bursts − spaced^{P}	3.39	3.35
Bursts: changing − fixed^{E}	1.73	2.86
No shape: bursts − spaced^{P}	0.80	0.05
Spaced: fixed − no shape^{P}	−1.02	−0.46
Spaced: fixed − changing^{E}	−0.02	0.02
Context about same − other noun^{E}	≤0.07	≤0.07

Table 2: Key comparisons in Experiment 2 (copy score, nats; seed-level means, 100 and 1,000 exposures pooled, and 1,000 alone). Intervals are in the text and Appendix I. The last row is the largest value over histories.

Bursts also give a noun a quarter as many training windows as spaced sentences. Fixed-shape bursts control for this: they have the same frame, presentation and number of windows, but their shape never changes, so memory can supply it. Changing-shape bursts scored 1.73 nats [1.25, 2.20] more than fixed-shape bursts^{E}, and 2.86 [1.99, 3.74] more at 1,000 exposures.

Recurrence on its own did little. No-shape bursts scored 0.80 nats [0.47, 1.13] more than no-shape spaced nouns^{P}, but the gain was confined to 100 exposures (1.55) and gone by 1,000 (0.05 [−0.30, 0.40]). Fixed-shape bursts also scored more than fixed-shape spaced nouns (1.69 [1.29, 2.08]^{P}), but at 100 exposures bursts also left the stored shape weaker (3.0 against 5.4 nats), so this difference cannot be attributed to recurrence; we had registered that it would be read only if stored knowledge was equal. At 1,000 exposures, when it was, the gain was 0.47 [0.27, 0.66]^{E}.

4.4 A stored shape does not itself reduce copying

We had predicted that fixed-shape nouns, presented spaced, would copy less than no-shape nouns, and they did, by 1.02 nats [0.76, 1.27]^{P}. But the comparison is confounded. No-shape nouns were trained in *A noun is a colour texture thing*, so at the test position they led the model to expect a colour word, whereas every other familiar noun led it to expect a shape (Appendix I). Fixed- and changing-shape nouns presented spaced share frame, presentation and expected word class, and differ only in whether one shape is stored. Their copy scores were the same (−0.02 [−0.16, 0.11]^{E}), within ±0.3 nats. A

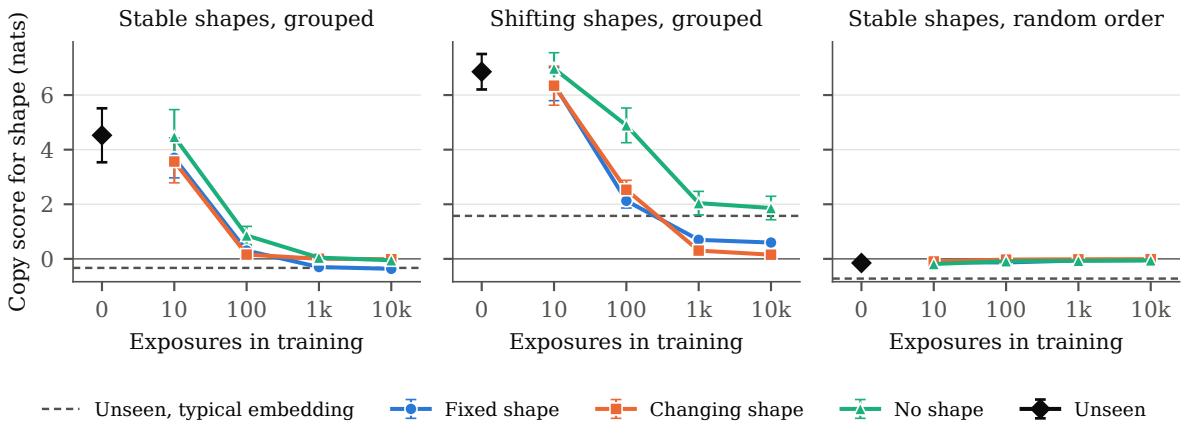


Figure 2: Experiment 1: copy score by number of exposures and noun history, all spaced (means over ten seeds, sizes averaged; 95% intervals). Diamond: unseen nouns. Dashed line: unseen nouns with a typical embedding.

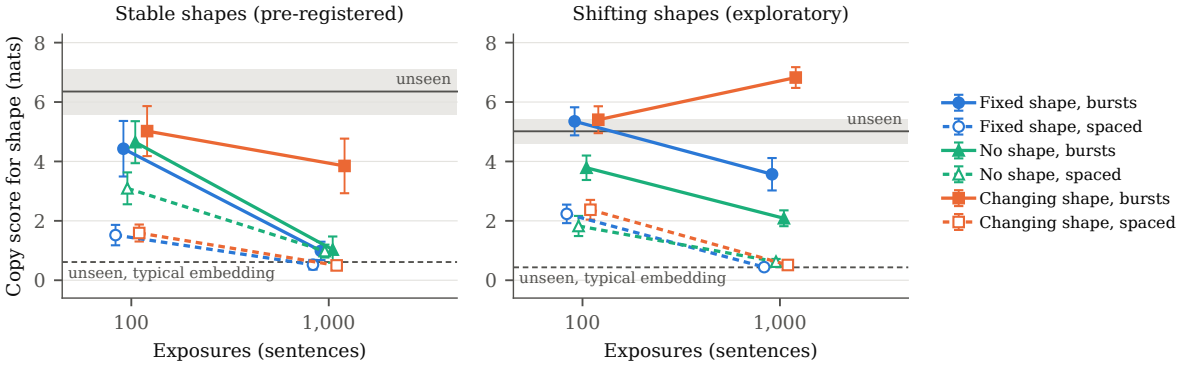


Figure 3: Experiment 2: copy score by history, presentation and exposure in stable-shape (left) and shifting-shape (right) corpora (means over ten seeds, sizes averaged; 95% intervals). Solid line and band: unseen nouns; dashed line: unseen nouns with a typical embedding.

stored shape mattered only together with bursts: fixed-shape bursts, whose earlier sentences predicted a shape that memory also supplied, lost copying, and changing-shape bursts did not.

4.5 What the model learns is a noun-specific switch, not a link

If changing-shape bursts taught the model to bind a noun to its current shape, copying should be weaker when the context sentence is about a different noun. It was not. In stable-shape models, the copy score with a context about the same noun exceeded the score with a context about another noun of the same history by at most 0.07 nats in Experiment 2, and 0.14 in Experiment 1^{E}. The model copied the most recent shape after the queried noun whichever noun the context described: the noun's history set whether it copied, not what it copied. A registered search for heads that carry the shape did not find them. Ablating the heads with the highest copying and prefix-matching scores, nearly all in the first layer, removed a mean of 22% of unseen nouns' copy score, against a predicted 50%, and no more than ablating random heads^{P} (Appendix H).

Shifting-shape corpora, where the trained kinds' own shapes can be recovered only from context, behaved differently (Figure 3, right^{E}). There, part of copying was tied to the noun (0.48 nats for unseen nouns, 1.45 for changing-shape bursts at 1,000 exposures), all bursts kept more copying (fixed shape 3.57, no shape 2.09 at 1,000, although bursts again weakened stored shapes), and changing-shape bursts copied more than unseen nouns (6.82 against 5.02). What a noun's history teaches therefore depends on what the rest of the corpus teaches about stored features.

4.6 Pretrained models

Finally, we asked whether the representation effect carries over to pretrained models, where real nouns

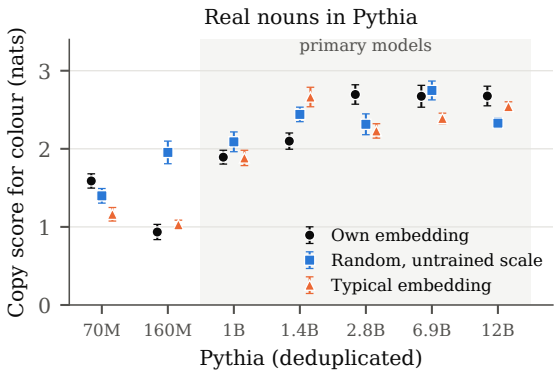


Figure 4: Copy score for colour of real nouns in Pythia models, with their own input embeddings or replaced ones (means over 44 nouns, 95% intervals).

have long histories. In Pythia models from 1B to 12B parameters (Biderman et al., 2023), we read 22 pairs of one-token nouns: one with a typical colour (*sky*, *leaf*, *bone*) and one without (*truck*, *hat*, *dog*), matched on frequency in the Pile (Gao et al., 2020; Liu et al., 2024) and on count or mass use. Each was read in frames such as *The noun was colour. The noun was*, and the copy score was computed over 11 colour words. The models did store typical colours: with no context, colour-typical nouns preferred their colour by 1.3 to 1.7 nats. Human colour norms (Paik et al., 2021) agreed with our coding for 15 of the 16 colour-typical nouns they cover.

Real nouns copied 1.9 to 2.7 nats of colour (Figure 4). We had predicted that replacing a noun's embedding with a random one of untrained scale would raise copying, more than replacing it with a typical one, and that colour-typical nouns would copy less. None of these held^{P}: the replacements raised copying in some models and lowered it in others, by up to 0.4 nats, and colour-typical and colour-variable nouns did not differ reliably. Most changes were also not small enough to count as zero, so the result is inconsistent rather than null. Changing only the size of a replaced embedding changed copying by less than 0.01 nats. The untrained-embedding effect of the small models did not transfer to these models and this probe.

5 Discussion

Which account? The memory account failed: a stored shape neither sped up the loss of copying nor, with word class matched, lowered it. The representation account explains why unseen nouns copy and why familiar nouns met one at a time stop, but not by itself why changing-shape bursts still copy after 1,000 exposures. The useful-context account explains all three. In our reading, the model learns for each noun whether the shape just mentioned has predicted the next one when memory could not. Changing-shape bursts are the only history in which that was true, and the only one in which copying lasted.

What is expected and what is not. That a model copies where copying lowers its loss is expected. Three things are not. Recurrence alone did not keep copying: no-shape bursts recur as often as changing-shape bursts but lost copying by 1,000 exposures. A stored shape did not compete with the context unless the context also repeated it. And the model never tied the copied shape to the noun, although in training the copied shape always belonged to that noun. What it learned is closer to a per-word switch on a general copying operation, set by the word's embedding. This is the population-level finding of Chan et al. (2022), that bursty items whose meaning changes favour learning in context, acting on single words within one model.

Implications. For probes with invented words, an unseen token's representation can produce most of an effect, so effects should be checked with representations that look familiar. For new tokens, initialising embeddings at the mean of existing ones (Hewitt, 2021) may suppress the very copying that lets a model learn them from context. For knowledge conflicts, our results suggest that entities whose attributes are stable within documents but vary across them may be the ones a model most readily takes from context. In Pythia we found no such link for colour, and testing it directly needs measures of each word's burstiness in pretraining.

Analogies with children. Two findings echo research on how children learn words, which we note as analogies only. Changing-shape bursts have the structure of the boxes in Dewar and Xu (2010): uniform within a box, varying between boxes. After one draw from a new box, 9-month-olds expected the next to match, an overhypothesis in the sense of Kemp et al. (2007). And at 100 exposures, spaced nouns stored their shape more strongly than nouns met in bursts, while bursts kept the context in use. Children likewise retain words and categories better from spaced than from massed exposures (Childers and Tomasello, 2002; Vlach et al., 2008), although they learn new labels better when the la-

bels are repeated in successive sentences (Schwab and Lew-Williams, 2016).

Limitations

The language is artificial, with 32 kinds, three features and a few sentence frames, and the models are small and trained with one recipe; most effects were largest in the smallest models. We tested copying only in the frame and feature that changing-shape nouns were trained on, with the context one sentence back. The other new nouns in a corpus may affect a noun's copying, and we did not train models with a single history. The head search found no specific heads, so the mechanism is unknown. Planned power for a 1-nat difference in Experiment 1 was 0.57, below the target of 0.8. The pretrained test used one model family, 22 noun pairs, three frames and colour only, and did not separate contexts that agree or conflict with a noun's typical colour. Earlier experiments in this language were registered in stages, some after related results were known (Appendix A).

Ethics statement

All training data are artificial; the pretrained models are public. No data were collected from people. Code, corpora, results and all trained models will be released; anonymised copies are provided as supplementary material.

Use of AI assistants

An AI assistant (Claude, Anthropic) was used to help design analyses, write code, review code and design, draft and edit text and check citations. It also reviewed the write-up of Experiment 1, which led to Experiment 2, and coded the colour typicality of the pretrained-model nouns in two independent passes. The authors are responsible for all content, including the accuracy of every citation.

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A Registration record

In the registrations, Study 1 is the earlier training experiments (Section 3; Appendices C and E), Study 2 an analysis of natural corpora and Study 3 an earlier test of pretrained models. Studies 2 and 3 do not bear on the claims of this paper; their methods and full results, and further results of Study 1, are in the long version in the supplementary material. Table A4 reports every registered hypothesis of all three studies.

All registrations are in one public project on the Open Science Framework (link withheld for review; an anonymised view-only link is in the supplementary material), and each states what was known when it was written. Table A1 lists them, Table A2 gives the label each registration uses for each row of Table A4, and Table A4 gives every registered hypothesis and its result.

Table A1: Registrations. Each was written before the models it concerns were trained or scored, or before the transcripts were read.

Registration	Written	Rows of Table A4	Known when it was written
Original	2 Mar 2026	Corpora, models and controls (long version)	Nothing trained
Amendment 1	7 Oct 2026	1, 2, 3, 6, 16	Random-order models trained and scored on an earlier test, with the old loader; no grouped model trained
Amendment 2	7 Oct 2026	7 (texture added as a co-primary test)	No grouped model trained
Amendment 3	8 Oct 2026	23, 24, 25	Study 1 results for rows 1 to 3, 6, 7 and 16 (old loader); no pretrained model scored
Amendment 4	8 Oct 2026	20, 21	Studies 1 and 3 results; no transcript read
Amendment 5	9 Oct 2026	4, 5, 8, 9, 17, 18, 19, 22	All earlier results (Study 1 with the old loader); four reviews of a draft; five exploratory checks, including the confound of the noun with another
Amendment 6	11 Oct 2026	11, 12, 13	Amendment 5 results (old loader); a prediction for the new models fitted to them
Amendment 7	12 Oct 2026	10, 14, 15	All results, with corrected scores
Amendment 8	15 Oct 2026	26 to 32	All results, with corrected scores; three rounds of reviews, which noted that fixed-file models memorised their files and that the novel-noun test was uninformative; no fresh-text model trained
Amendment 9	23 Oct 2026	Familiarity study (Appendix H)	All results above; no familiarity corpus built
Amendment 10	24 Oct 2026	Recurrence, frames, rows and pretrained models (Appendix I)	Amendment 9 results and a blind review of them; no Amendment 10 corpus used, model trained or pretrained model run on its items

Table A2: Rows of Table A4 and the labels used in the registrations.

Row	Label	Row	Label	Row	Label	Row	Label
1	H4b	9	H13b	17	H14a	25	H8
2	H1b	10	H22	18	H14b	26	H23
3	H2b	11	H17	19	H15	27	H24
4	H11	12	H18	20	H9	28	H25
5	H12	13	H19	21	H10	29	H26
6	H5b colour	14	H20	22	H16	30	H27
7	H5b texture	15	H21	23	H6	31	H28
8	H13a	16	H3b	24	H7	32	H29

The loading correction. The model code ties the input embedding to the output layer. On the graphics

card used for training (through DirectML) the tie was lost, so every Study 1 model was trained and saved with two separate matrices. The first version of the scoring code rebuilt each model with the tie in place before loading its weights, so the output matrix overwrote the input embedding. We found this on 12 October 2026, fixed the loader, and rescored all 315 Study 1 models then trained, 461 intermediate checkpoints of the random and grouped Regular models and the trained-noun items, with the same items. A check on eight randomly chosen models confirmed the fix: mean loss on each model's own training corpus was 0.07 to 0.16 nats per token with the corrected loader, close to the last logged training loss (0.08 to 0.23; dropout is off at test), and 0.58 to 1.97 with the old one. Studies 2 and 3 were not affected. The old scores are kept in the repository.

Table A3: Study 1 registered tests before and after the correction (shape-over-colour index, nats). Rows 10, 14 and 15 were registered and scored after the correction.

Row	Old loader	Corrected	Verdict before → after
1	1.36 vs 0.18	6.49 vs 0.47	Supported → supported
2	0.18	0.47	Not supported → not supported
3	0.18 vs 0.09	0.47 vs −0.25	Not supported → supported
4	difference 1.45	difference 6.93	Supported → supported
5	difference 1.17	difference 2.76	Supported → supported
6	1.36 vs 0.80	6.49 vs 2.12	Supported → supported
7	1.10 vs −0.04	5.36 vs 0.46	Supported → supported
8	slope 0.29	slope 1.63	Supported → supported
9	0.24 vs 0.18	1.43 vs 0.47	Not supported → supported
11	0.21 vs 0.18	0.50 vs 0.47	Not supported → not supported
12	0.29 vs 0.24	0.80 vs 1.43	Not supported → not supported (reversed)
13	0.21 vs −0.08	0.50 vs 0.10	Supported → supported
16	difference 0.0007	difference 0.005	Equivalent to zero → equivalent to zero

Table A4
All registered hypotheses and results; registered secondary analyses are reported in the text. Rows 1 to 16 concern fixed-file models and rows 26 to 32 fresh-text models. Seed-level tests average the three sizes within each of five seeds (t-based 95% intervals, two-sided; tests one-sided unless marked, 4 degrees of freedom), so a one-sided test can be below .05 while the two-sided interval includes zero (row 13). The registered p is the test named in the registration, one-sided unless marked, Holm-corrected within its registration; for rows 26 to 32 it is the seed-level test, corrected within rows 26 to 29 and within rows 30 to 32. Corpus intervals resample sources and then documents; Pythia and BabyLM intervals resample nouns.

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
	1. Does grouping teach the fixed feature?					
1	Grouped order gives a higher index than random order	6.49 vs 0.47	6.02 [5.24, 6.79]	$< .001$	$< .001$	Supported
2	Random order gives an index above zero	0.47	0.47 [−0.61, 1.56]	.15	.061	Not supported
3	Random order: Regular above Mixed-slot	0.47 vs −0.25	0.73 [−0.69, 2.15]	.11	.032	Supported by the registered test, not at the seed level
4	Grouped: Regular above No stable feature	6.49 vs −0.44	6.93 [5.47, 8.40]	$< .001$	$< .001$	Supported
5	Grouped: Colour stable below No stable feature	−3.20 vs −0.44	2.76 [1.99, 3.52]	$< .001$	$< .001$	Supported
6	Grouped: Regular above Mixed-slot, shape over colour	6.49 vs 2.12	4.37 [1.98, 6.76]	.004	$< .001$	Supported
7	Grouped: Regular above Mixed-slot, shape over texture	5.36 vs 0.46	4.89 [2.69, 7.10]	.002	$< .001$	Supported
8	The index rises with log B across six orders	Slope 1.63 (SE 0.10)	1.63 [1.46, 1.80]	$< .001$	$< .001$	Supported
9	B = 3.6 above random order	1.43 vs 0.47	0.96 [0.63, 1.28]	$< .001$	$< .001$	Supported
10	The same, Objects corpora, different-object runs	1.18 vs 0.22	0.96 [0.37, 1.54]	.005	$< .001$	Supported
26	Fresh text: grouped order gives a higher index than random order	9.52 vs −1.56	11.08 [9.57, 12.59]	$< .001$	$< .001$	Supported
27	Fresh text: B = 3.5 above random order	1.28 vs −1.56	2.84 [1.41, 4.27]	.003	.005	Supported
28	Fresh text, grouped: Regular above No stable feature	9.52 vs −3.98	13.50 [11.48, 15.52]	$< .001$	$< .001$	Supported
	2. What kind of grouping helps?					
11	512-token window above 128, random order	0.50 vs 0.47	0.03 [−0.44, 0.50]	.44	.87	Not supported
12	512-token window above 128, B = 3.6	0.80 vs 1.43	−0.63 [−0.90, −0.37]	.003 (two-sided)	1.00	Not supported; reliably reversed
13	512-token window: Regular above No stable feature	0.50 vs 0.10	0.40 [−0.04, 0.85]	.032	.045	Supported; weak
14	Grouped: different-object above same-object runs	5.65 vs 2.99	2.66 [1.57, 3.75]	.001	$< .001$	Supported
15	B = 3.6: different-object above same-object runs	1.18 vs 0.45	0.73 [0.15, 1.32]	.013	$< .001$	Supported
29	Fresh text, grouped: different-object above same-object runs	7.60 vs 5.87	1.72 [0.54, 2.90]	.008	.008	Supported
	3. Is the expectation tied to the noun?					
16	Fixed files, random order: index higher after the same noun (novel nouns)	Difference 0.005; equivalent to zero (p = .001), but the novel nouns were untrained (Section 3)			.46	Not supported
30	Shape reassigned, grouped: noun-specific index above zero, novel nouns	0.37	0.37 [0.24, 0.49]	$< .001$	.001	Supported

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
31	The same, above grouped Regular (fresh text)	0.37 vs 0.12	0.25 [0.10, 0.39]	.005	.005	Supported
32	Shape reassigned, grouped: noun-specific index above zero, nouns from training	11.24	11.24 [10.44, 12.03]	$< .001$	$< .001$	Supported
17	Pythia: noun-specific index rises with size (wording matched)	$\rho = -.42$			1.00	Not supported
18	Pythia 410M to 12B: noun-specific index above zero (wording matched)	−0.15 (SE 0.12)			1.00	Not supported
19	Pythia: 410M above 160M across ten training seeds 4. What do natural input and pretrained models show?	0.15 vs −0.15			.012	Supported
20	Speech to children: B above 1	3.57 [3.13, 4.30]			$< .001$	Supported
21	Speech to children above written text	3.57 vs 1.43; difference 2.14 [1.55, 2.91]			$< .001$	Supported
22	Speech to children differs from adult conversation (two-sided)	3.57 vs 2.33; difference 1.24 [0.51, 2.03]			$< .001$	Supported; higher
23	Pythia: index rises with size	$\rho = .92$			.003	Supported
24	Pythia: index rises over training	Mean within-size $\rho = -.34$			1.00	Not supported
25	BabyLM baselines (10M, 100M words): index above zero	0.47 [−1.05, 1.93]; 0.37 [−0.04, 0.73]			.52; .12	Not supported

Fresh text and shape reassignment. Rows 26 to 32 were registered on 15 October 2026, after reviews of the manuscript had noted that fixed-file models memorised their files and that the test with novel nouns was uninformative, and before any fresh-text model was trained. The corpora were built, and their checksums registered, before training. Amendment 8 names the conditions by their folder names (for example reassigned_grouped).

B Deviations from the registrations

1. Two registered analyses are not reported here: the Bayesian reference model of Amendment 1, which is in the repository, and the prediction of Amendment 6 for the 512-token models, which had been fitted to scores later corrected (Appendix A).

2. Added after registration: the seed-level tests for rows 1 to 3 and 6 to 8, the seed-level intervals, and intervals for the differences between natural corpora (from the registered two-stage bootstrap). Effective clustering was computed before it was registered as the scale for a figure of the long version and is reported there as exploratory.

3. The two models of about 70 billion parameters were scored with Apple's MLX library, because PyTorch crashed the machine; on smaller models the two differed by 0.003 to 0.010 nats on average.

4. In the hand coding of Study 2, a rule for labels not used for a physical object was added during coding and applied by both coders, and one pair of the reliability subset was discussed with an AI assistant and left out of the agreement check.

Smaller technical changes (software versions, download addresses, a stale file in one public model repository) are listed in the repository.

C Model and training settings

Table C1: AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up then cosine decay, dropout 0.1, untied input and output embeddings. Fixed-file models read windows starting at random positions in the corpus; fresh-text models read non-overlapping windows, each at most once, in random order (the 3.4M and 25.6M models used about 60% of the windows available, the 11.0M models 94 to 96%). Byte-pair tokeniser with 351 entries for the Regular corpora and 314 to 339 for the other corpus families; every artificial word is one token. Seeds 42, 123, 456, 789 and 1001 each set the corpus, the initial weights and the batch order. These five seeds are those of the earlier experiments; Experiments 1 and 2 used ten (adding 2718, 3141, 1618, 1414 and 1732), untied embeddings by design, a tokeniser of 329 entries and weights saved every 1,000 steps. All models are pre-LayerNorm decoders with learned absolute position embeddings, GELU feed-forward layers and a final LayerNorm. Checkpoints were saved every 500 steps. Training ran on one Windows PC through DirectML.

Setting	3.4M	11.0M	25.6M
Layers, heads, width	4, 4, 256	6, 8, 384	8, 8, 512
Feed-forward width	1,024	1,536	2,048
Steps	5,000	8,000	10,000
Batch (128-token windows)	32	32	16
Batch (512-token windows)	8	8	4
Peak learning rate	3×10^{-4}	2×10^{-4}	1×10^{-4}
Warm-up steps	200	400	500

Table C2: Memorisation. Mean next-token loss (nats per token) on 300 windows from each model's own training text and from a new arrangement of the same sentences (100 further passes, 27 for the Objects corpora, never used in training); 15 models per cell. At the seed level, the gap between the two losses was larger for fixed-file than for fresh-text models by 0.57 to 1.16 nats, with every interval excluding zero.

Corpus	Fixed file: own file	Fixed file: new arrangement	Fresh text: own text	Fresh text: new arrangement
Regular, random	0.11	1.25	0.55	0.55
Regular, B 3.6 (fresh: 3.5)	0.10	1.26	0.54	0.54
Regular, grouped	0.09	0.85	0.35	0.35
No stable feature, grouped	0.09	1.00	0.39	0.39
Objects, same-object runs, grouped	0.08	0.66	0.28	0.28
Objects, different-object runs, grouped	0.09	0.76	0.32	0.32

D Data and code

All code, corpora, test items, model scores and analysis outputs, and the final weights and training logs of every trained model, will be released on publication; anonymised copies of the code and results are in the supplementary material. A README gives the command that regenerates each table and figure, with MD5 checksums for the fixed inputs. The pretrained models are public at the revisions listed.

E Language, corpora and earlier measures

This appendix describes the artificial language and the measures of the earlier experiments.

E.1 Corpora

Each corpus describes 32 kinds of object, each named by a novel noun (*wug*, *dax*, *blorp* and so on), with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each a novel word from a list of ten per dimension, and is described by one sentence in one of five frames:

1. *A wug is a [shape] [colour] [texture] thing.*
2. *Look at the wug. It is [shape] and [colour].*
3. *Here is another wug. This one is [shape].*
4. *The wug I showed you is [shape], but a different [colour].*

5. *That is a wug because it is [shape].*

Shape is named in all five frames, colour in three and texture in one. The frames are the same in every corpus we compare, so they cannot by themselves explain a difference between corpora, although they may interact with what is manipulated. The 448 sentences of a corpus, each used once, make up one pass. Five corpora were generated for each condition, one per seed.

Content. The corpora differ in which feature, if any, stays fixed within a kind (long version, Table 1). In **Regular** corpora every kind keeps one shape across its 14 exemplars, while colour and texture vary. In **Mixed-slot** corpora each kind keeps one word from the shape list, but for about half the kinds that word appears in the texture slot, and colour varies; no single dimension is fixed across all kinds. In **No stable feature** corpora, built from Regular, each exemplar's shape is drawn at random from the list, so nothing is fixed within a kind. In **Colour stable** corpora, also built from Regular, shape is drawn in the same way and each kind keeps one colour instead. Frame 4 is unchanged, so in Colour stable corpora it names the kind's fixed colour after the words *a different*.

Objects. A further set of corpora separates two kinds of nearby repetition. In the Regular corpora, two sentences about a kind always describe different exemplars. In speech, nearby mentions often return to the same object. The Objects corpora keep the Regular content but give each kind 13 objects and describe each object four times, once in each of frames 1, 2, 3 and 5. Frame 4 is left out because it states that the colour differs. A pass of these corpora has 1,664 sentences.

Order. The same sentences were arranged in several orders. In **random** order every pass is shuffled. In **grouped** order, within each pass, the sentences about each kind are shuffled, cut into runs of four, and the runs are shuffled across kinds. Every sentence occurs exactly as often as in random order. In **graded** orders each run is kept with a set probability and otherwise split into single sentences before shuffling, which gives intermediate levels of clustering. In the Objects corpora, a **same-object run** is the four descriptions of one object, and a **different-object run** is four different objects of one kind, one per frame. A Latin square over objects and frames ensures that every sentence is used once per pass in every order. Same-object and different-object runs were each arranged fully grouped and at an intermediate level.

Reassigned. In every corpus so far, a kind's fixed feature is fixed for the whole corpus, so the noun alone predicts it, and a model can predict the shape in a sentence from its noun without comparing the sentence with the one before. In Reassigned corpora, built from Regular, each kind receives a new shape, drawn at random from the ten, in every pass, and keeps it in all of its sentences in that pass. The next sentence about the same kind still repeats the shape, but the noun no longer says which shape it will be. These corpora were arranged in random and grouped order.

Fresh and fixed text. A model trained many times over on one short file can memorise it. Our main models were therefore trained on fresh text: 7,500 passes of a corpus (2,100 for the Objects corpora), each arranged with its own random draw by the rule of its order, 34 to 35 million tokens in all. Training read this text in non-overlapping 128-token windows, so no window was seen twice. A second set of models was trained in the common way, on one fixed file of 11,648 sentences (26 passes, or 7 for the Objects corpora; about 120,000 tokens), with windows starting at random positions, so that each position was seen 160 to 290 times. These fixed-file models memorised their files. Their loss was 0.08 to 0.11 nats per token on their own file but 0.66 to 1.26 on a new arrangement of the same sentences, while fresh-text models scored 0.28 to 0.55 on both (Table C2). Table 1 of the long version shows which conditions were trained on fresh text and which on a fixed file; where both exist, we report fresh text first. The fresh graded order, built by the same rule as the fixed one at B = 3.6, measured B = 3.5.

Clustering. We measure clustering with one statistic, B, here and in Study 2. Take the sequence of mentions of object nouns in a text, and every pair of consecutive mentions that are no more than 100 words apart. P is the share of these pairs that name the same kind. P_0 is the same share after the sentences (or utterances) of each document are shuffled, averaged over shuffles. B = P/P_0, pooled over documents. For example, in *Look at the dog. The dog is big. Where is the cup? Here is another dog.*, the consecutive pairs are dog–dog, dog–cup and cup–dog, so P = 1/3. B = 1 means no clustering beyond what the content predicts. Random-order Regular corpora have B = 0.9, slightly below 1, because each pass is shuffled

separately: within a pass, a sentence's kind has 13 other members among 447 sentences, against 363 among 11,647 when the whole corpus is shuffled. The graded levels (long version, Table 1) were chosen after the clustering of natural input had been measured (Study 2): 1.4 and 3.6 match B in written text (1.43) and in speech to children (3.57). They were fixed from the corpora alone before any graded-order model was trained.

Table E1: Study 1 corpora. Every corpus describes 32 kinds. Fresh: 7,500 passes (2,100 for Objects), each arranged anew and read once in non-overlapping 128-token windows. Fixed: one file of 11,648 sentences (10,304 to 12,992 for three further controls: No kind label, Paraphrased and Weak label), read in 128-token windows at random positions unless marked. Each order of each corpus was trained at three sizes with five seeds (15 models). The further controls are reported in the long version.

Corpus	What stays fixed within a kind	Orders (measured B)	Text	Models
Regular	Shape. Colour and texture vary.	Random (0.9); graded (3.5); grouped (23)	Fresh	45
No stable feature	Nothing	Grouped (23)	Fresh	15
Objects	Shape; each object described four times	Same-object runs (24); different-object runs (24)	Fresh	30
Reassigned	Shape within a pass; a new shape for each kind in every pass	Random (0.9); grouped (23)	Fresh	30
Regular	Shape	Random (0.9); graded (1.4, 3.6, 7, 12); grouped (23)	Fixed	90
Regular	Shape	Random (0.9); graded (3.6)	Fixed, 512-token window	30
Mixed-slot	One shape-list word, in the shape slot for about half the kinds and the texture slot for the rest	Random; grouped	Fixed	30
No stable feature	Nothing	Random; grouped; random with a 512-token window	Fixed	45
Colour stable	Colour. Shape varies.	Random; grouped	Fixed	30
Objects	Shape; each object described four times	Random (1.0); same-object runs (3.6, 24); different-object runs (3.6, 24)	Fixed	75
Further controls: no kind label, weak label, paraphrased, frequency matched, added noise, feature swap	As Regular, with one change each	Random	Fixed	90
Total				510

E.2 Models and training

We trained decoder-only transformers of three sizes: 3.4 million parameters (4 layers, 4 heads, width 256), 11.0 million (6 layers, 8 heads, width 384) and 25.6 million (8 layers, 8 heads, width 512). They were trained for 5,000, 8,000 and 10,000 steps with batches of 32, 32 and 16, peak learning rates of 3×10^{-4}, 2×10^{-4} and 1×10^{-4}, AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up, cosine decay and dropout of 0.1. Each training example is a 128-token window, which spans several sentences, so attention crosses sentence boundaries. The tokeniser for the Regular corpora has 351 entries (314 to 339 for the other corpus families), and every artificial word is a single token. A seed sets the corpus, the initial weights and the batch order (the long version lists the models per condition). For the window test, models were trained with 512-token windows and a quarter of the batch size, so that every step saw the same number of tokens. All models have separate input and output embeddings, and the parameter counts include both; the model code ties the two, but the tie was lost on the training hardware (Appendix A). Table C1 lists every setting.

E.3 Measuring the expectation

Eight novel nouns that never occur in training (*snerg*, *twick* and so on) are used in test items. They are in the vocabulary, so each is one token, but their input embeddings were never trained; weight decay shrinks them by less than 10% over training, so they remain distinct random vectors. With the intended tie, the same vectors would have served as the novel nouns' output weights and would have changed in training; we cannot say whether the models would then have tracked these nouns. Novel nouns keep learned knowledge about a noun out of the test, but the test then depends on the model registering that the same untrained token recurs. We therefore also test with nouns from training, whose embeddings are trained (long version). Each item gives the model one sentence in frame 1 describing an exemplar, followed by two continuations that differ in one word:

> Context: *A snerg is a krannish pulek skorli thing.* Repeat: *A snerg is a* ***krannish*** *temil tramel thing.* New: *A snerg is a* ***tova*** *temil tramel thing.*

In this item the probed dimension is shape. The repeat continuation keeps the context's shape, the new one does not, and both give new values for the other two dimensions. The model scores each continuation by the sum of the log probabilities of its tokens given the context, so the difference, Δ, includes the changed word and its effect on the words after it. Δ is the model's preference for repeating the probed dimension, in nats. Trained models generally prefer words they have just seen, so Δ alone mixes the dimension with copying. Our index is the difference between dimensions:

$$\text{index} = \Delta_{\text{shape}} - \Delta_{\text{colour}}$$

Copying that affects shape and colour equally cancels. Copying that differs between dimensions does not, which is one reason every conclusion rests on comparisons between corpora tested in the same way, and why we report Δ for each dimension as well. The shape-over-texture index is defined in the same way. With one exemplar, nothing in the context shows which dimension is fixed, so differences between trained conditions come from training.

Every item is also given with a different novel noun in the continuations (*A yeff is a krannish temil tramel thing*). The **noun-specific index** is the index after the same noun minus the index after a different noun. A kind-tied expectation scores above zero on it; a noun-blind one scores zero.

With 25 draws of features per noun, three probed dimensions and two continuation nouns there are 1,200 one-exemplar items. A further 2,400 items with two or four exemplars test the use of evidence in the context (long version). Untrained models of the same architecture, with the Regular tokeniser, give an index of 0.02 nats (SD 0.07).

E.4 Statistics

Each model gives one index, its mean over items. Models of the three sizes trained with the same seed share a corpus, so the five seeds are the units of replication. We average the three sizes within each seed and compare conditions by paired t tests over the five seed means (4 degrees of freedom); the differences and 95% intervals given in the text are these seed-level estimates. Registered contrasts were tested one-sided, with Holm (1979) correction within each registered family. For fresh-text models the registered test was this seed-level test. For fixed-file models it was a linear mixed model (index $\sim$ condition + size, with a random intercept for seed) or, for some contrasts, a Welch t test over models; the long version gives the seed-level results beside them. Single conditions were compared with zero by one-sample t tests. Results reported with a Holm-corrected p, or called registered, were registered; all others are exploratory. We registered 0.2 nats as the smallest effect of interest and tested equivalence to zero against it with two one-sided tests (Lakens, 2017). This bound is small next to the effects reported here, so the equivalence tests are strict.

F Test-noun representation and changed-word scoring

These analyses were added after the registrations and are exploratory. They rescore the 120 fresh-text models on the one-exemplar items (`scripts/recut_rescore.py`,

`scripts/recut_rescore_analysis.py`). **Typical tokens**: each novel noun's input embedding is replaced by the mean input embedding of the 32 training nouns plus $r\,u_j$, where r is the mean distance of the training-noun embeddings from that mean and the u_j are orthonormal random directions (fixed seed), so the eight tokens look like ordinary nouns but stay distinct; output weights are unchanged. **Changed-word** Δ: the difference in log probability of the one word that differs between the continuations, given the identical prefix. With the original embeddings and whole-continuation Δ, the rescoring reproduced the reported scores (144,000 item scores; largest difference 2×10^{-5} nats). Trained-noun values are from the registered trained-noun items (long version in the supplementary material; `scripts/recut_noun_store_check.py`).

Table F1: Index and noun-specific index under four scorings (nats; mean of seed means). Original: unseen tokens with untrained embeddings, as reported; typical: the same tokens with typical embeddings; full: whole-continuation Δ, as reported; word: changed word only.

	Index				Noun-specific index			
Condition	orig./full	typ./full	orig./word	typ./word	orig./full	typ./full	orig./word	typ./word
Regular, random	−1.56	−1.54	−1.26	−1.50	−0.05	−0.31	0.01	−0.20
Regular, graded (3.5)	1.28	−1.41	1.45	−1.25	−0.03	−0.15	0.03	−0.04
Regular, grouped	9.52	1.46	9.65	1.38	0.12	0.00	0.12	0.01
No stable feature, grouped	−3.98	−3.51	−3.74	−3.24	−0.35	−0.26	−0.34	−0.21
Objects, same-object runs	5.87	−1.85	6.06	−1.87	−0.07	0.14	−0.06	0.06
Objects, different-object runs	7.60	−1.25	7.59	−1.33	−0.01	0.19	−0.01	0.11
Reassigned, random	0.21	0.18	0.32	0.27	0.02	−0.10	0.03	−0.01
Reassigned, grouped	9.80	3.87	9.94	3.98	0.37	0.70	0.37	0.74

Table F2: Preference for repeating the context's shape and colour after one exemplar, same noun, for the three kinds of test noun (nats; whole-continuation Δ).

	Δ shape			Δ colour		
Condition	Unseen	Typical	Trained	Unseen	Typical	Trained
Regular, random	−1.55	−1.52	0.03	0.01	0.02	0.05
Regular, graded (3.5)	1.18	−1.52	0.06	−0.10	−0.12	0.19
Regular, grouped	6.88	−0.72	0.06	−2.64	−2.18	−1.94
No stable feature, grouped	−3.88	−3.51	−3.58	0.10	0.00	0.17
Objects, same-object runs	6.87	−1.17	−0.09	0.99	0.68	0.46
Objects, different-object runs	8.05	−1.14	0.04	0.46	0.11	0.01
Reassigned, random	0.22	0.24	0.24	0.02	0.06	0.10
Reassigned, grouped	6.81	1.17	10.01	−2.99	−2.70	−1.65

Table F3: Noun-familiarity curve: unseen tokens' input embeddings blended from their untrained values (0) to typical embeddings (1). Index and Δ shape, same noun (nats; mean of seed means); slope per unit blend with 95% interval over five seeds.

Condition	Measure	0	0.25	0.5	0.75	1	Slope
Regular, random	Index	−1.56	−1.83	−1.93	−1.79	−1.54	0.03 [−0.22, 0.28]
Regular, random	Δ shape	−1.55	−1.83	−1.92	−1.78	−1.52	0.05 [−0.26, 0.35]
Regular, grouped	Index	9.52	6.70	4.32	2.10	1.46	−8.28 [−9.87, −6.70]
Regular, grouped	Δ shape	6.88	4.27	2.08	−0.08	−0.72	−7.82 [−9.59, −6.04]
Objects, same-object runs	Index	5.87	3.71	1.08	−1.41	−1.85	−8.23 [−10.05, −6.40]
Objects, same-object runs	Δ shape	6.87	4.59	1.90	−0.65	−1.17	−8.52 [−10.22, −6.83]
Objects, different-object runs	Index	7.60	4.53	1.87	−0.82	−1.25	−9.22 [−9.97, −8.46]
Objects, different-object runs	Δ shape	8.05	4.85	2.08	−0.69	−1.14	−9.57 [−10.14, −9.00]
Reassigned, grouped	Index	9.80	8.03	5.34	3.87	3.87	−6.41 [−8.74, −4.08]
Reassigned, grouped	Δ shape	6.81	4.83	2.24	0.97	1.17	−6.06 [−7.28, −4.84]

Mean input-embedding norms: trained nouns 0.55, unseen tokens 0.36, typical tokens 0.55.

G Terms, hypotheses and registrations

Terms. The registrations and the detailed appendices that follow use the labels of the registration documents. Table G1 gives the correspondence with the terms of the main text.

Table G1: Main-text terms and registration terms.

Main text	Registrations and Appendices H and I
Experiment 1	Amendment 9 (the familiarity study)
Experiment 2	Amendment 10, Part A
Row analysis (Section 4.1)	Amendment 10, Part C
Pretrained models (Section 4.6)	Amendment 10, Part B
Stable-shape corpora	Regular (grouped) corpora
Shifting-shape corpora	Reassigned (grouped) corpora
Spaced, bursts	singles, runs
Changing shape	uninformative (Experiment 1), rotating (Experiment 2)
No shape	shape-absent
Copy score	carry
Typical embedding	typical row
Random embedding of untrained scale	atypical_init

Hypotheses. Table G2 lists every registered hypothesis of Experiments 1 and 2, the row analysis and the pretrained test, the claim of the main text it bears on, and its outcome under the registered decision rules (Holm correction within each experiment; direction in at least two of three model sizes; for the pretrained test, Holm within each model and support in at least four of five models). Equivalence bounds were set in each registration: ± 1 nat in Amendment 9, where the pilot gave seed-level standard deviations near 0.8 nats, and ± 0.3 nats in Amendment 10, the smallest history effect seen in Amendment 9. The hypotheses of the earlier experiments are in Appendix A.

Table G2: Registered hypotheses and outcomes. Full statistics: Tables H1 (Experiment 1), I1 (Experiment 2), I8 (row analysis) and I9 (pretrained models).

Label	Prediction (main-text terms)	Outcome
Experiment 1 (Amendment 9)		
M1	Fixed-shape nouns store their shape (knowledge $>$ 0.2 nats at 1,000 exposures, rising)	Passed in all three corpora
H30	Copying falls faster with exposure for fixed-shape than no-shape nouns	Not supported (wrong direction)
H31	No-shape nouns (1,000 and 10,000) differ from unseen nouns with a typical embedding; equivalence ± 1	Different (0.33) and equivalent
H32	Fixed-shape nouns' preference for their own shape rises with exposure	Supported
H33	Ablating candidate heads removes more than half of unseen nouns' copying, more than random heads	Not supported
H34	Ablating candidate heads lowers trained kinds' copying in shifting-shape corpora	Supported; random heads did the same
H35	Candidate heads' attention falls faster for fixed-shape nouns	Not tested (required H33)
R1	Unseen nouns copy more in grouped than random order	Replicated
Experiment 2 (Amendment 10, Part A); stable-shape corpora		
M2	Unseen nouns copy more than 1 nat	Passed
K1	Fixed-shape nouns' knowledge equal (± 1) in bursts and spaced	Not equivalent: H41 not interpreted
H40	No-shape: bursts differ from spaced; equivalence ± 0.3	Bursts higher
H41	Fixed shape: bursts differ from spaced; equivalence ± 0.3	Bursts higher (not interpreted)
H42	Changing shape: bursts copy more than spaced	Supported
H43	Spaced: fixed shape copies less than no shape	Supported; confounded by expected word class (Section 4)
H44	Spaced no-shape nouns differ from unseen nouns with a typical embedding; equivalence ± 0.3	Higher; not equivalent
Row analysis (Amendment 10, Part C); Experiment 1 models		
H49	Scaling unseen nouns' rows to the trained kinds' norm lowers copying	Supported (driven by the 3.4M models)
Pretrained models (Amendment 10, Part B)		
H45	A random embedding of untrained scale raises copying over the own embedding	Not supported (2 of 5 models)
H46	The same embedding raises copying over a typical embedding	Not supported (2 of 5 models)
H47	Colour-typical nouns copy less than colour-variable nouns	Not supported (0 of 5)
H48	The H45 increase differs between colour-typical and colour-variable nouns	Inconclusive

H The familiarity study (Amendment 9)

Registration. The design, hypotheses and analysis were registered before any corpus was built (Amendment 9). An addendum, written the same day before any corpus was built, fixed details the registration left open (36 context draws per noun; uniform frame proportions; exposures placed at evenly spaced read positions from 5% to 95% of training, after warm-up for every size; random in-window position; the 40 nonce nouns; how the pilot fixes the number of nouns per cell; how a hypothesis counts as supported; the gate for H35; and that heads without prefix matching are described as copying heads). A pilot (one seed, 3.4M models, never used in a test) set three nouns per cell; with the registered rule the expected power for a 1-nat difference was 0.57, below the planned 0.8, which we report as a shortfall. Corpus checksums were recorded before training and matched on the training machine; a checker verified every exposure, window and frame from the tokens. All models were trained on one machine.

Design. In every model, 40 nonce nouns fill 13 cells (three per cell, four unseen): fixed shape, uninformative shape (a new shape in every sentence, drawn from the corpus's shape distribution) and shape-absent (a sixth frame, *The noun is colour and texture*, also used by trained kinds in one sentence in fifteen), each at 10, 100, 1,000 and 10,000 exposures, plus unseen. Nouns rotate over cells across ten seeds. Each familiarity sentence lies wholly inside one 128-token window, at most one per window, between runs in grouped corpora. Base corpora: Regular grouped, Regular random, Reassigned grouped; sizes 3.4M, 11.0M and 25.6M; ten seeds; 90 models with untied embeddings. Measures are read from

the full log-softmax over the ten shape words at the position that predicts the shape in *A noun is a*: carry $= L(s) - \bar{L}$ after a context sentence with shape s; stored preference $= L(s^*) - \bar{L}$; knowledge $= L(s^*)$ minus the mean over the other nine shapes after an unrelated sentence, where $\bar{L}$ is the mean over shapes other than s and s^*.

Table H1: Registered tests of the familiarity study (Regular grouped models unless stated; seed-level estimates over ten seeds, sizes averaged within seed; nats, proportions for H33a, attention probability for H35). Holm p is adjusted over H30 to H34; H35 was not entered because H33(a) failed. Sizes: number of the three model sizes with the registered direction.

	Test	Estimate	95% CI	Holm p	Sizes	Verdict
M1	Knowledge at 1,000 exposures, fixed shape (each corpus)	9.53; 8.61; 8.26				Passed
H30	Carry falls faster with exposure for fixed-shape than shape-absent nouns (slope difference)	0.16	[0.01, 0.30]	1.000	0	Not supported
H31	Shape-absent (1k and 10k) minus unseen with typical embedding (two-sided; TOST ± 1)	0.33	[0.13, 0.52]	.013	3	Different, and equivalent within 1 nat
H32	Stored preference rises with exposure, fixed shape (slope)	2.97	[2.79, 3.16]	$<.001$	3	Supported
H33a	Ablating candidates removes more than half of unseen carry (proportion)	0.22	[0.07, 0.36]	1.000	2	Not supported
H33b	Random heads minus candidates, unseen carry after ablation	0.64	[−0.43, 1.71]	1.000	2	Not supported
H34	Reassigned: trained-kind carry intact minus candidates ablated	7.41	[6.02, 8.80]	$<.001$	3	Supported
H35	Attention slope difference, fixed shape minus shape-absent	0.002	[0.000, 0.005]	–	2	Not tested (gate)
R1	Replication: grouped minus random Regular, unseen carry	4.68	[3.71, 5.65]	$<.001^a$	3	Replicated

[a] Outside the Holm family (replication); unadjusted one-sided p. M1 values: Regular grouped; Regular random; Reassigned grouped (all with slopes above zero, $p < .001$). H31 TOST $p < .001$.

Table H2: Carry (nats) by noun type and exposure, mean of seed values (sizes averaged). Unseen: never in training; typical: the unseen nouns with a typical embedding; trained: the 32 base kinds.

Corpus	Unseen	Typical	Trained	Familiar nouns	10	100	1k	10k
Regular grouped	4.53	−0.33	−0.02	Fixed shape	3.70	0.30	−0.30	−0.37
				Uninformative	3.57	0.16	0.00	−0.01
				Shape-absent	4.47	0.86	0.05	−0.05
Reassigned grouped	6.85	1.57	9.98	Fixed shape	6.40	2.13	0.70	0.60
				Uninformative	6.34	2.53	0.30	0.15
				Shape-absent	6.97	4.89	2.04	1.86
Regular random	−0.15	−0.72	−0.10	Fixed shape	−0.13	−0.13	−0.06	−0.05
				Uninformative	−0.10	−0.03	−0.02	−0.01
				Shape-absent	−0.19	−0.09	−0.07	−0.05

Table H3: Knowledge and stored preference (nats) in Regular grouped models, mean of seed values.

Type	Measure	10	100	1k	10k
Fixed shape	Knowledge	1.59	8.02	9.53	10.61
	Stored	1.41	7.74	9.56	10.71
Uninformative	Knowledge	0.04	−0.03	0.00	0.03
	Stored	0.01	−0.03	−0.01	0.03
Shape-absent	Knowledge	−0.14	−0.17	0.12	0.08
	Stored	0.11	−0.16	0.10	0.06

Table H4: Head analysis by corpus and size (means over ten seeds). Candidates were the k heads with the highest product of prefix-matching and copying scores; in grouped models they were in layer 0 in all but one model (a 3.4M Regular grouped model chose one layer-1 head), and in random-order models a few were in later layers. In the 3.4M and 11.0M models layer 0 has exactly k non-candidate heads, so the random draws repeat one set. Prefix: highest prefix-matching score of any head (uniform attention gives about 0.027). Carry of unseen nouns and of trained kinds intact, with the candidates mean-ablated, and with random heads ablated (mean of 200 or 20 draws). DLA: change in unseen carry predicted by the candidates' direct paths.

				Unseen carry			Trained carry			
Corpus	Size	k	Prefix	Intact	Cand.	Random	Intact	Cand.	Random	DLA
Regular grouped	3.4M	2	0.028	8.10	−0.42	5.26	−0.25	−0.85	−0.71	−0.04
	11.0M	4	0.037	2.98	5.80	1.95	0.08	0.17	−0.40	−0.01
	25.6M	6	0.037	2.50	0.73	0.81	0.12	−0.60	−0.18	−0.01
Reassigned grouped	3.4M	2	0.029	6.48	4.22	3.27	9.59	2.83	3.37	−0.12
	11.0M	4	0.032	7.21	6.56	4.82	10.10	5.02	2.45	−0.03
	25.6M	6	0.031	6.88	0.26	3.26	10.26	−0.11	3.39	−0.01
Regular random	3.4M	2	0.028	−0.24	−0.09	−0.34	−0.08	0.00	−0.11	−0.01
	11.0M	4	0.030	0.04	0.25	0.03	−0.06	0.04	−0.11	−0.01
	25.6M	6	0.030	−0.25	0.00	−0.37	−0.18	−0.10	−0.27	−0.01

Table H5: Attention of the candidate heads from the position that predicts the shape to the context's shape word (mean over candidates; mean of seed values; exploratory).

Corpus	Type	Unseen	10	100	1k	10k
Regular grouped	Fixed shape	0.155	0.144	0.082	0.071	0.071
	Uninformative		0.147	0.088	0.076	0.074
	Shape-absent		0.154	0.097	0.077	0.075
Reassigned grouped	Fixed shape	0.219	0.217	0.108	0.055	0.059
	Uninformative		0.217	0.157	0.071	0.070
	Shape-absent		0.219	0.170	0.115	0.139
Regular random	Fixed shape	0.089	0.088	0.076	0.067	0.067
	Uninformative		0.087	0.082	0.075	0.073
	Shape-absent		0.089	0.084	0.075	0.073

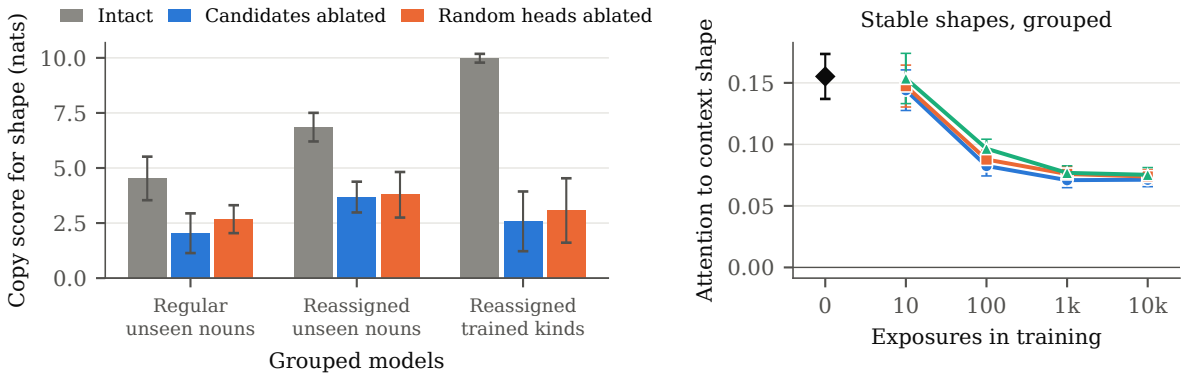


Figure H1: Experiment 1. Left: copy score intact, with the candidate heads mean-ablated, and with random heads ablated (nats; mean over ten seeds and three sizes, 95% intervals over seeds). Right: the candidates' attention from the position that predicts the shape to the context's shape word, stable-shape grouped models (exploratory).

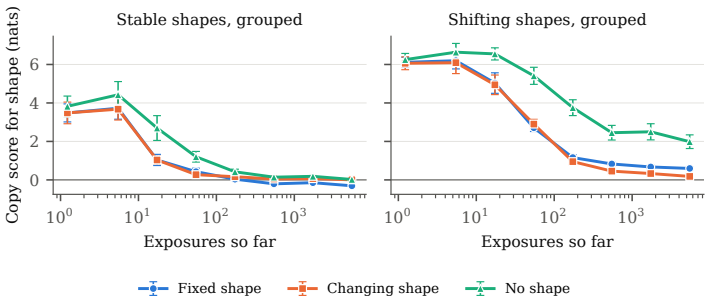


Figure H2: Experiment 1. Copy score against the exposures each noun had received by each saved checkpoint (exploratory; nats; mean over ten seeds of size-averaged values, 95% intervals). Exposure is confounded with training time.

I Recurrence, matched frames, embedding rows and pretrained models (Amendment 10)

Registration. Amendment 10 has three parts: Part A is Experiment 2, Part B Experiment 3 (pretrained models) and Part C the row analysis. It was registered after the Amendment 9 results, after its corpora were built and checksummed, and before any of them was used, any of its models was trained, or any pretrained model was run on its items. It answered two confounds raised by a blind review of the Amendment 9 results (by an AI assistant given only the write-up of Experiment 1): familiarity nouns never recurred within a window, and shape-absent nouns never appeared in a frame with a shape slot after *is a*. The noun lists for Part B were fixed, with checksums, before the pilot. A pilot on Pythia-410M (never in a test) gave a mean increase of 0.13 nats from the original to the atypical_init row, so the smallest effect of interest for H48 was 0.30 nats by the registered rule. One deviation: some Pythia tokenizers define no padding token, so the scoring script falls back to token id 0; padded positions are masked and only the last real token is read, and batched and single-item scores agreed exactly. No score had been produced by the affected models.

Part A design. Everything not stated is as in Amendment 9 (Appendix H). Histories: *fixed*, frame 1 (*A noun is a shape colour texture thing*) with the noun's own shape; *absent*, frame 7 (*A noun is a colour texture thing*), which names no shape; *rotating*, frame 1 with a shape drawn from the corpus's shape distribution, a new one for every sentence when presented singly and one per run, changing between runs, when presented in runs. Presentation: *single*, one sentence per unit; *run*, four consecutive sentences about the noun in one window. Exposures (100 or 1,000) count sentences, so runs and singles are matched in sentences, not in units or spacing. Trained kinds use frame 7 in one sentence in fifteen, so the frame is not new. Three nouns per cell (12 cells) and four unseen nouns; Regular grouped (confirmatory) and Reassigned grouped (exploratory) base corpora; three sizes and ten seeds (60 models). A checker re-read every corpus and verified exposures, units, frames, shapes within runs and window placement; all 40 corpora matched checksums recorded before training.

Part C (embedding rows in the Amendment 9 models). In the 60 final grouped models of Amendment 9 we replaced the input rows of unseen nouns and scored carry again; the unmodified condition reproduced the Amendment 9 scores. Conditions: own row; own direction scaled to the mean norm of the trained kinds' rows (ownnorm); a random direction at that norm (randnorm); the trained kinds' mean row. Shape-absent nouns at 10,000 exposures were also scored with their own direction scaled to the unseen nouns' mean norm (shrunk).

Part B (pretrained models). 44 nouns in 22 pairs, matched on Pile frequency and count/mass type, were coded for colour typicality by two independent AI passes with a fixed prompt before any model was run (exact agreement 94%, weighted $\kappa = 0.96$); nouns that are also colour terms or that both passes flagged as polysemous were excluded. Colour-typical nouns are mostly natural kinds, so kind was a covariate. Each noun was read in three neutral frames with each of the eleven basic colour terms that are one token; carry is the mean over frames and ordered colour pairs of $\log p(c_1 \mid c_1) - \log p(c_1 \mid c_2)$, with colour probabilities renormalised over the colour set. Only the noun's input row was changed.

Table I1: Registered tests of Part A (Regular grouped models; 100 and 1,000 exposures pooled; seed-level estimates over ten seeds, sizes averaged within seed; nats). Holm p is adjusted over H40 to H44. Sizes: number of the three sizes with the estimate's direction. Readings follow the registered table: "above" means the Holm threshold was met and the direction held at two of three sizes; TOST p for H40, H41 and H44 was above .99, so none was equivalent.

	Test	Estimate	95% CI	Holm p	Sizes	Reading
M2	Unseen carry minus 1 nat (manipulation check)	5.35	[4.59, 6.12]	$<.001^a$		Passed
K1	Knowledge, fixed runs minus fixed singles (equivalence ±1)	−1.09	[−1.45, −0.73]	$<.001^a$		Not equivalent (H41 not interpreted)
H40	Absent runs minus absent singles (two-sided; TOST ±0.3)	0.80	[0.47, 1.13]	$<.001$	3	Above
H41	Fixed runs minus fixed singles (two-sided; TOST ±0.3)	1.69	[1.29, 2.08]	$<.001$	3	Above (not interpreted)
H42	Rotating runs minus rotating singles (one-sided)	3.39	[2.69, 4.09]	$<.001$	3	Supported
H43	Fixed singles minus absent singles (one-sided)	−1.02	[−1.27, −0.76]	$<.001$	3	Supported
H44	Absent singles minus unseen with typical row (two-sided; TOST ±0.3)	1.42	[1.12, 1.73]	$<.001$	3	Above

[a] Outside the Holm family; unadjusted p. The M2 estimate is unseen carry minus 1 nat, so unseen carry was 6.35 nats.

Table I2: Carry (nats) by cell, mean of seed values (sizes averaged) with 95% intervals over ten seeds. Unseen nouns have no exposures; the typical row replaces an unseen noun's input row with one typical of trained nouns.

Corpora	Noun	100 exposures	1,000 exposures
Stable shapes	Fixed shape, spaced	1.52 [1.18, 1.86]	0.52 [0.36, 0.69]
	Fixed shape, bursts	4.43 [3.49, 5.36]	0.99 [0.68, 1.30]
	No shape, spaced	3.09 [2.56, 3.63]	0.98 [0.76, 1.20]
	No shape, bursts	4.65 [3.94, 5.36]	1.03 [0.59, 1.47]
	Changing shape, spaced	1.59 [1.30, 1.88]	0.50 [0.38, 0.62]
	Changing shape, bursts	5.02 [4.18, 5.86]	3.85 [2.93, 4.77]
	Unseen	6.35 [5.59, 7.12]	
	Unseen, typical embedding	0.61 [0.28, 0.95]	
Shifting shapes	Fixed shape, spaced	2.23 [1.92, 2.55]	0.44 [0.37, 0.51]
	Fixed shape, bursts	5.35 [4.88, 5.82]	3.57 [3.02, 4.11]
	No shape, spaced	1.83 [1.49, 2.17]	0.63 [0.53, 0.72]
	No shape, bursts	3.79 [3.38, 4.20]	2.09 [1.82, 2.35]
	Changing shape, spaced	2.37 [2.04, 2.71]	0.51 [0.38, 0.64]
	Changing shape, bursts	5.40 [4.95, 5.86]	6.82 [6.48, 7.17]
	Unseen	5.02 [4.61, 5.42]	
	Unseen, typical embedding	0.43 [0.32, 0.55]	

Table I3: Knowledge of the noun's own shape and expectation of a shape rather than a colour at the query position (slot: log of the summed probability of the ten shapes minus that of the ten colours), Regular grouped models, mean of seed values (exploratory). Unseen nouns: knowledge −0.04, slot 1.49.

	Knowledge		Slot	
Noun	100	1,000	100	1,000
Fixed shape, singles	5.37	7.80	4.03	6.16
Fixed shape, runs	3.02	7.98	3.28	6.24
Shape-absent, singles	0.00	−0.09	−2.98	−4.93
Shape-absent, runs	0.05	−0.25	−1.12	−4.43
Rotating shape, singles	−0.08	0.01	5.04	5.30
Rotating shape, runs	0.14	0.06	3.74	5.24

Table I4: The Part A contrasts in Reassigned grouped models (exploratory; seed-level, unadjusted two-sided p).

	Contrast	Estimate	95% CI	p
H40	Absent runs minus absent singles	1.71	[1.36, 2.06]	<.001
H41	Fixed runs minus fixed singles	3.12	[2.71, 3.53]	<.001
H42	Rotating runs minus rotating singles	4.67	[4.28, 5.06]	<.001
H43	Fixed singles minus absent singles	0.11	[−0.07, 0.29]	.196

Table I5: Matched contrasts in Part A (exploratory; seed-level, nats, 95% intervals). Rotating minus fixed runs: same frame, presentation and number of occasions; the two differ only in whether the shape changes between runs (and so in whether a shape is stored). Fixed minus rotating singles: same frame, presentation and expectation of a shape at the test position (Table I3); they differ only in whether the noun has one stored shape.

Base	Contrast	Pooled	100	1,000
Regular grouped	Rotating runs − fixed runs	1.73 [1.25, 2.20]	0.59 [0.23, 0.95]	2.86 [1.99, 3.74]
	Fixed singles − rotating singles	−0.02 [−0.16, 0.11]	−0.07 [−0.29, 0.15]	0.02 [−0.04, 0.08]
	Fixed runs − fixed singles	1.69 [1.29, 2.08]	2.91 [2.21, 3.60]	0.47 [0.27, 0.66]
Reassigned grouped	Rotating runs − fixed runs	1.65 [1.35, 1.96]	0.05 [−0.05, 0.16]	3.26 [2.64, 3.87]
	Fixed singles − rotating singles	−0.10 [−0.23, 0.02]	−0.14 [−0.34, 0.06]	−0.07 [−0.20, 0.05]
	Fixed runs − fixed singles	3.12 [2.71, 3.53]	3.12 [2.61, 3.62]	3.13 [2.59, 3.66]

Knowledge, fixed runs minus fixed singles (Regular grouped): −2.35 [−2.82, −1.88] at 100 exposures, 0.18 [−0.18, 0.54] at 1,000.

Table I6: Noun-specific carry (exploratory): carry when the context sentence is about the query noun minus carry when it is about another noun of the same cell, with the same shape (nats; seed-level means). Near zero means the model copies the context's shape whichever noun the context was about.

		Regular grouped		Reassigned grouped	
Experiment	Noun	100	1,000	100	1,000
2	Unseen	0.05		0.48	
	Fixed, singles	0.01	−0.02	0.22	0.06
	Fixed, runs	0.06	−0.05	0.58	0.65
	Shape-absent, singles	0.07	0.05	0.25	0.05
	Shape-absent, runs	0.01	0.00	0.46	0.70
	Rotating, singles	0.04	0.01	0.35	0.05
	Rotating, runs	0.05	0.05	0.58	1.45
1	Unseen	0.14		0.45	
	Fixed shape	−0.05	−0.02	0.21	0.01
	Uninformative	0.00	0.00	0.51	0.04
	Shape-absent	0.05	0.00	0.45	0.22

Table I7: Part A by model size (Regular grouped; seed means, 100 and 1,000 exposures pooled; nats).

Size	Unseen	Typical row	H40	H41	H42	H43	H44	Rot. − fixed runs
3.4M	7.67	−0.30	0.65	2.41	5.05	−0.84	1.67	2.94
11.0M	5.74	0.64	1.23	1.46	2.82	−0.78	1.25	1.24
25.6M	5.65	1.51	0.53	1.20	2.29	−1.43	1.36	1.00

Table I8: Part C: carry (nats) by input row, mean over seeds (sizes averaged). H49 (registered, Regular grouped): ownnorm minus own −0.52 [−0.85, −0.20], one-sided p = .003, direction at 2 of 3 sizes, supported. In Reassigned models the same contrast went the other way (0.27 [0.12, 0.41]; exploratory). Randnorm minus ownnorm: 0.09 (Regular), 0.03 (Reassigned).

	Unseen nouns				Shape-absent, 10k	
Corpus	Own	Ownnorm	Randnorm	Trained mean	Own	Shrunk
Regular grouped	4.53	4.00	4.09	−0.44	−0.05	0.53
Reassigned grouped	6.85	7.12	7.15	1.05	1.86	4.16

Table I9: Part B registered tests by primary model (nouns or pairs as units; Holm within model over H45 to H48). H45: atypical_init minus original; H46: atypical_init minus typical row; H47: colour-typical minus colour-variable nouns, original rows; H48: difference between groups in the H45 increase (TOST against ± 0.30). A hypothesis needed Holm in at least four of the five models. Outcome: H45 and H46 met Holm in two of five models, H47 in none, and H48 was inconclusive (equivalent in one model).

	H45		H46		H47		H48	
Model	Est.	Holm	Est.	Holm	Est.	Holm	Est.	TOST p
1B	0.20	met	0.21	met	0.00	–	0.07	.097
1.4B	0.34	met	−0.22	–	−0.01	–	−0.08	.021
2.8B	−0.38	–	0.08	–	−0.16	–	0.11	.222
6.9B	0.07	–	0.36	met	−0.09	–	0.10	.127
12B	−0.35	–	−0.21	–	0.11	–	0.08	.066

Mixed model over the five primary models (carry $\sim$ condition $\times$ group + model + kind; random intercepts for noun and pair, random condition slopes by noun): atypical_init 0.00 (p = .956), typical row −0.04 (p = .559), colour-variable group 0.06 (p = .527).

Table I10: Part B: carry of the context's colour (nats; mean over nouns) by model and input row, and with original rows by group. Atypical: the atypical_init direction at the noun's own norm (exploratory). 70M and 160M are descriptive. Prior entropy: entropy of the colour distribution after *The noun was* with no context (nats; uniform over eleven colours is 2.40).

Model	Original	Atypical_init	Atypical	Typical row	Colour-typical	Colour-variable	Prior entropy
70M	1.59	1.40	1.41	1.16	1.61	1.56	2.09
160M	0.93	1.95	1.95	1.03	0.87	1.00	2.01
1B	1.89	2.09	2.09	1.88	1.89	1.89	1.95
1.4B	2.10	2.44	2.44	2.66	2.10	2.10	1.97
2.8B	2.70	2.31	2.31	2.23	2.61	2.78	1.92
6.9B	2.67	2.75	2.75	2.39	2.63	2.72	1.94
12B	2.68	2.33	2.33	2.54	2.73	2.62	1.91

Table I11: Part B checks (exploratory). Stored colour: with no context, log probability of a colour-typical noun's coded colour minus the mean over the other colours after *The noun was* (nats; mean over 22 nouns, and the share of nouns above zero). TOST: equivalence of the H45 and H46 differences within ± 0.30 nats over nouns. Norm only: atypical minus atypical_init (same direction, different norm).

Model	Stored colour	Share > 0	H45 TOST p	H46 TOST p	Norm only
70M	0.72	0.73	.059	.129	0.014
160M	0.90	0.77	1.000	1.000	0.000
1B	1.28	0.82	.125	.091	0.001
1.4B	1.30	0.91	.736	.172	0.000
2.8B	1.55	0.86	.776	.001	0.000
6.9B	1.66	0.95	.012	.801	−0.002
12B	1.73	0.86	.740	.023	−0.001

Table I12: Part B, Pythia-1.4B deduplicated across training (exploratory): carry (nats; mean over nouns) by training step and input row.

Row	1,000	2,000	4,000	8,000	16,000	32,000	64,000	143,000
Original	0.73	1.09	0.72	1.97	1.46	1.67	2.39	2.10
Atypical_init	0.81	1.45	1.10	2.08	1.82	2.11	2.56	2.44
Typical row	0.72	1.06	0.60	1.64	1.17	1.70	2.70	2.66